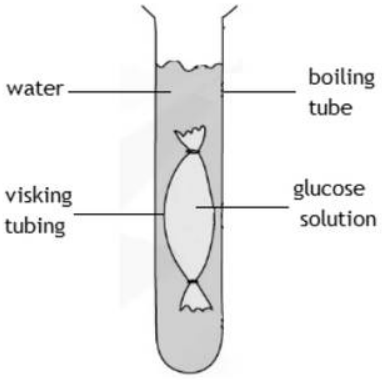
Type MS · Explain diffusion in a context and factors affecting its rate

Answer 1 of 7 · Question 5 part a · 3 marks

2025 JUN · Paper 02 (regional R) · find part a in this mark scheme extract

5	(a)	• greater concentration gradient/high concentration in small intestine; • (more) glucose collides/in contact with, wall/epithelium, of small intestine; • faster rate of diffusion;	allow reverse argument	3
	(b) (i)	Benedict's (solution);		1
	(ii)	(heated) water bath/Bunsen burner/measuring cylinder;		1
	(iii)	• diagram; • labels;;  	1 mark for each set of two correct labels. ignore any inclusion of a water bath in diagram accept sugar solution and glucose/sugar solution	3
	(c)` (i)	villus/villi		1
	(ii)	any three from • many/large number of villi/have microvilli; • increase surface area; • thin walls; • for shorter diffusion distance; • capillaries; • to maintain diffusion/concentration gradient;	explanation must be linked to feature	3
				Total for Question 5 = 12 marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 2 of 7 · Question 2 part b · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

2 (a)	(i)	<table><tr><th>Part</th><th>Name of part</th><th>Function</th></tr><tr><td>X</td><td>mitochondria;</td><td></td></tr><tr><td>Y</td><td></td><td>controls activities of the cell/stores DNA / genetic information;</td></tr></table>		1 1
	(ii)	**A,** more cell structures can be seen with greater resolution		1
	(iii)	60mm/6cm; 60 ÷ 0.05; 1200;	Ecf two marks for correct calculation from incorrect measurement Full marks for correct final answer	3
(b)		• diffusion; • from a high concentration (in the cell) to a lower concentration (in the blood)/down a concentration gradient;		1 1

Total question 2 = 8 marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 2 of 7 · Question 2 part b · 2 marks

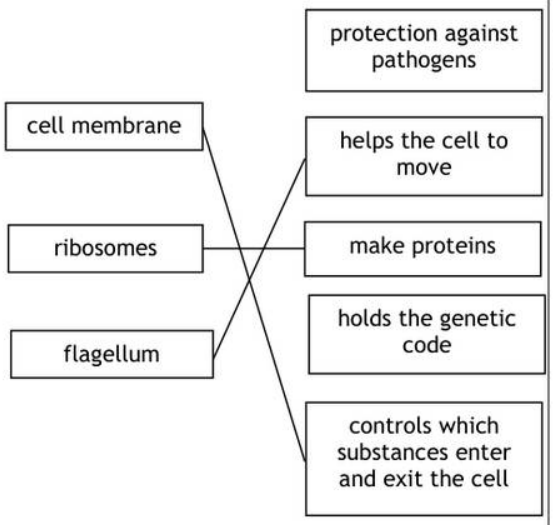
2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 3 of 7 · Question 3 part b.i · 1 mark

2020 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

3 (a) (i)		Reject more than one line from each structure One mark for each correct line	3
(ii)	Any two from: • (bacterial cell) has a cell wall (1) • (bacterial cell) does not have a nucleus (1) • (bacterial cell) has a plasmid (1)		2
(b) (i)	**The only correct answer is B diffusion** *A is incorrect because it is down a diffusion gradient* *C is incorrect because it is not removal from the bacterium* *D is incorrect because osmosis only involves movement of water*		1
(ii)	• glucose (1) • carbon dioxide (1)	Answers must be in the correct order Allow correct chemical formulae	2
(iii)	Any two from: • to provide energy (1) • for movement (1) • for growth/reproduction/divide/mitosis (1) • for chemical reactions/producing organelles/named organelle (1)		2

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 3 of 7 · Question 3 part b.i · 1 mark

2020 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Total for Question 3 = 10 marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 4 of 7 · Question 6 part c · 2 marks

2024 JUN · Paper 01 · find part c in this mark scheme extract

6 (a) (i)	3 x 3 x 6; 54 (cm^2)	full marks for correct final answer ecf	(2)
(ii)	3 x 3 x 3; 27 (cm^3)	full marks for correct final answer ecf	(2)
(iii)	54:27; 2:1	ecf for wrong values	(2)
(b)	Any five from • red blood cells contain haemoglobin; • to transport oxygen/binds to oxygen; • blood plasma; • contains dissolved substances; • named substance e.g. nutrients/hormones/CO_2/heat; • transported through blood vessels/named blood vessel;		(5)
(c)	• diffusion; • (nutrients) higher in concentration outside the cell than inside the cell/down a concentration gradient/from high to low concentration;		(2)

Total for Question 6 = 13 marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 5 of 7 · Question 3 part a.iv · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

Question number					

3	(a)	(i)	surface area = 24 (mm^2);
			volume = 8 (mm^3);
			3:1;
				Full marks for correct final answer	3

		(ii)	C 92;		1
		(iii)	movement from a high concentration to a low concentration;		1
		(iv)	• reduces the time taken to become colourless/rate of diffusion/reaction is faster; • the diffusion distance is shorter/greater area over which diffusion/collisions can occur;		2
	(b)	(i)	• use acid heated to different temperatures/idea of different temperatures; • compare/measure the time taken to become colourless/compare rate of diffusion;		2
		(ii)	Any two from: • the side length of the cube/size of the cube/surface area/volume of cube; • acid concentration; • concentration of alkali/indicator in the cube;	accept volume of acid Allow volume of indicator	2

Total 11 marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 5 of 7 · Question 3 part a.iv · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 6 of 7 · Question 5 part a.iii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

5 (a)	(i)	• flows from artery; • where blood is at high pressure; • into vein; • because blood is at a low pressure;	mp1- accept in context of artery leaving heart mp3- accept in context vein of returning to the heart If comparative answer given for pressure score marking point 2 and 4	4
	(ii)	glomerulus;		1
	(iii)	• large (surface) area; • for diffusion / molecules/ named molecules to pass out(of blood);	More area for diffusion = 2 marks	2
	(iv)	more urea;		1
(b)		• no/less gradient between liquid X and blood (glucose)/blood and X have same concentration/eq; • glucose does not pass out/ glucose is not lost: • by diffusion • needed for respiration; • minimises osmotic effect;		3
(c)		• protein digested/broken down into <u>amino acids</u>; • (excess) **<u>amino acids</u>** broken down/deaminated; • urea; • less protein means less urea produced; • less urea means less to be filtered/removed ; • high concentration of urea/nitrogenous waste can be toxic;	MP3– accept urea from protein breakdown	4
				Total =15

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 6 of 7 · Question 5 part a.iii · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 7 of 7 · Question 7 part a.ii · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

7 (a) (i)	Any two from the following • causes cancer; • gene mutation; • incorrect protein made/shape of protein incorrect;		2
(ii)	• changing the way free radicals behave; • to allow normal flow of substances/named substance in and out of cells;		2
(iii)	• 40g = 60% of daily requirement; • $\frac{40 \times 100}{60}$; • = 67g;	Allow 66.7g	3
(b)	• add fresh orange juice to a test tube/named suitable container; • add DCPIP drop by drop until it no longer decolourises; • record the volume of DCPIP added; • repeat with the same volume of boiled orange juice; • compare volumes;		5
			Total 12

Type MS · Explain diffusion in a context and factors affecting its rate

Answer 7 of 7 · Question 7 part a.ii · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks